ENEL 585 Final Project

Fall 2026 | Common requirements

Oct. 1, 2026 · Common requirements for Projects A–E

Design and experimentally validate a controller for your chosen converter. Equipment is provided; your group assembles and verifies external wiring, establishes operating limits, designs and tests the controller in simulation and HIL, and compares predictions with bench measurements.

Prepare through remote HIL: begin model and interface checks before Bench 1 and complete controller testing before Bench 2. Ungraded HIL sign-off is required before closed-loop hardware operation.

Project scope

Form a group of up to four in your lab section and choose A–E; projects may repeat. B–E need no approval. A is limited to two groups per section: email the instructor; the first two requests per section are confirmed and can start immediately. Choices are fixed at proposal submission. Groups sharing a station across sections must each verify its wiring. B–E normally use single-phase banks; a three-phase bank may be assigned if needed. Use the resistance settings and connections issued for your assigned bank when choosing proposal operating points.

Read your station sheet with this handout, the Inverter Box Setup Guide and PLECS Setup Guide. Figures omit sensing and gate drives; C–E show breaker/precharge paths. Crossings without dots are not connections.

Provided	Your responsibility
Converter equipment, instruments and controller hardware	Connection diagram, external wiring and verification, operating targets and protection settings
HIL plant models (power stage, sensing and I/O setup; C–E transformer voltage and precharge), with no control; remote access. C–E: example PLECS synchronization (PLL) model	Switched PLECS model of your station, controller design, firmware generated with PLECS Coder, HIL model configuration and remote validation
Station sheet: circuit, equipment and hardware limits	Operating points, switching frequency, bench measurements and interpretation

Deliverables and dates

Deliverable	Course weight	Due or scheduled time
Proposal	4%	Fri Oct 9
Oral design review	8%	Bench 1: Oct 30 (B01), Nov 6 (B02)
Group demonstration	10%	Registrar-scheduled examination slot
Individual questions	8%	During the demonstration
Final report	10%	Fri Dec 11

Total: 40% of the course grade. Remote HIL readiness and bench milestones require sign-off but carry no separate marks.

1 Engineering requirements

Student wiring and shared stations

Each group prepares its own connection diagram. The first group assembles the external wiring; the second traces and verifies it. Make only justified, approved changes and record the configuration you leave. Both groups must explain and verify the station and obtain approval before energizing.

Operating limits and targets

Establish limits, probe settings, ADC scaling and protection from the station sheet, Setup Guide and component datasheets, including IGBT rating conditions. Explain the AC-filter role, tolerances and resonance. Compare simulated and measured peak/RMS filter currents, including ripple, with ratings. A separate AC-filter switching-ripple derivation is not required; calculate ripple where it supports a design choice, such as boost-inductor peak current or single-phase DC-link ripple.

Choose at least two feasible operating points (E: your inductive and capacitive reactive-power commands) with measurable targets. State them and your protection thresholds in the proposal; they are reviewed before you energize. Any later change is re-reviewed and carries no penalty when justified; report the original and revised values with the reason.

Switching frequency and controller design

Choose and justify a switching frequency at or below the 10 kHz lab ceiling, imposed by losses in the teaching inverter. Address losses, bandwidth, sampling delay and relevant resonance; include ripple/peak-current calculations only where needed for the design. Frequency and protection settings require approval before powered switching.

For each loop, derive the small-signal plant, design the compensator and predict crossover and phase margin. Include sampling/PWM rates, delay, limiters and anti-windup where needed.

C–E synchronize to the 60 Hz transformer using the provided example PLECS model of a synchronization (PLL) block, which outputs the transformer-voltage angle; include it in your controller. C/D build their sinusoidal current reference from that angle, not from a scaled copy of the measured voltage; E sets its current phase from it for its P/Q commands. Show lock in simulation and HIL. A/B are standalone and need no synchronization. Each project has two differential voltage probes; and two current probes. Read A/B DC input from the ITECH display.

Simulation and implementation

Compare switched and linearised PLECS models where the small-signal approximation applies. Generate LAUNCHXL-F28P55X firmware using PLECS Coder; preserve control logic and document settings through simulation, HIL and bench testing.

Your diagram must show power, gating, sensing/scaling, control loops, protection and instruments, including physical-to-ADC and PWM-to-switch mapping.

Measurements and interpretation

Demonstrate safe, stable closed-loop operation at each operating point and every test, signal and metric on your station sheet. Compare with predictions and explain limitations. Additional bench transients and efficiency measurements are optional. Define each metric, including how periodic ripple is treated. For C/D, phase error is the angle between the fundamentals of the transformer voltage and input current; distinguish it from true input power factor, and state the THD harmonic range and measurement bandwidth.

Estimate main losses from datasheets to justify the operating points and switching frequency. Explain discrepancies between simulation results and experimental results using evidence.

2 Remote HIL and laboratory work

Prepare remotely in stages

Remote HIL access and models are being prepared; instructions will be posted on D2L when ready. Before Bench 1, check model parameters and I/O mapping, and attempt controller communication. Full closed-loop validation is not required yet.

Before Bench 2, complete the HIL checklist: startup, stable operating points, every station-sheet transient, C–E synchronization lock, relevant limits and protective shutdown. Further tests are optional.

Ungraded HIL readiness check

At the start of bench 2, demonstrate the checklist on HIL to the instructor or TA before closed-loop hardware operation. Identify versions and settings and explain the response. Earlier sign-off is welcome.

Staff record ungraded sign-off; no separate submission or advance deadline applies. Repeat failed checks before progressing and recheck material changes. Retain evidence for the report and demonstration.

HIL does not replace checks of physical wiring, sensor polarity and hardware protection.

Three bench sessions

Session	B01	B02	Required progress
1 Bring-up	Oct 30	Nov 6	Design review; assemble or trace and verify external wiring; sensing, gating and protection checks; approved low-power bring-up.
2 Closed loop	Nov 20	Nov 27	HIL check and A's control-design review at the start; after sign-off, establish stable closed-loop operation at reduced power.
3 Measurements	Dec 4	Dec 1 10:00–12:15	Demonstrate your operating points and every station-sheet test; capture the required waveforms and metrics.

Catch-up uses scheduled bench time, reducing data collection and the evidence available for assessment. No routine extra time is provided. Report difficulties promptly; excused absences follow course policy.

Why precharge is required

Diodes rectify AC with all gates off, so closing the supply onto a discharged DC link through inductance can overshoot: an ideal undamped series LC circuit, initially unenergized, charges its capacitor to twice an applied DC step. Actual peaks depend on source, circuit and damping; do not rely on supply OVP. On C–E, B_2 applies AC through R_{pc} with B_{bp} open; close B_{bp} once the DC link has settled. B_1 (16 A primary) de-energizes the transformer.

Connect the ITECH only to approved A/B DC-bus terminals, never C–E or a node that can exceed its voltage limit. Filter energy can return during transients; use the approved mode, limits and shutdown procedure.

Be ready to answer: With gating disabled, where does charging current flow, and what changes when the precharge bypass breaker closes?

Bench checks

Before energizing, obtain instructor/TA confirmation of wiring, probe scaling, gate mapping and protection. Independently verify shared wiring against your diagram. Measure below 5 V on the DC link and external capacitors before rewiring; switch auxiliary power off before handling gate BNCs. Interpret error inputs and LEDs using the Setup Guide sections Protection Related and Status Check.

Common startup and shutdown

Standard single-phase banks have 20 switched 250 Ω parallel branches (R = 250/n Ω for n closed switches); use the issued configuration and resistance settings for an assigned three-phase bank. The hardware limits on your station sheet are equipment ceilings, not operating targets. Commanded DC voltage is limited to 60 V, except C/D at 70 V. Your operating points and trip thresholds must keep peak and RMS stresses, including ripple and transients, within the component ratings; they are reviewed with your protection settings.

Use the sequence for your station below with your reviewed settings. Begin at a reduced-power starting point and increase in steps. Use only the intended bank switches for live load changes; select a reproducible change within your reviewed limits. On C–E, the instructor or TA runs your first precharge and energization with you and confirms the station controls and expected settled DC-link voltage.

ITECH supply for A and B

1. With PWM inhibited and the AC load connected, set the ITECH to source mode, output off and 0 V, with the current limit at 10 A and overvoltage protection at 60 V. Leave the station transformer (used by C–E) disconnected from the inverter and load.

2. After sign-off, enable the source and raise DC voltage in steps, to at most 58 V, while monitoring voltage and charging current. Enable PWM only after charging has settled, then approach your first operating point.

3. For normal shutdown, reduce demand, inhibit PWM, ramp the ITECH to zero and switch its output off. Ramping down pulls the DC link down with it; confirm the ITECH display reads below 5 V before handling. The filter capacitors discharge through their bleeders.

Transformer supply for C D and E

1. Identify the 16 A primary breaker B_1, the 20 A secondary breaker B_2 and the 16 A bypass breaker B_{bp}. With B_2 open and the DC link below 5 V, open all DC-load switches, inhibit PWM and open B_{bp}. Confirm the charging loop: 60 Ω through two resistors for C/D; one 30 Ω resistor with neutral return for E.

2. Close B_2 to apply AC through R_{pc}. Watch the DC-link voltage rise and settle near the expected unloaded value, roughly the input peak less diode drops. If it fails to rise, stalls or behaves abnormally, open B_2 and involve the instructor or TA.

3. Close B_{bp} before connecting load or enabling PWM; never operate the converter through R_{pc}.

4. Connect the starting load and enable switching in your reviewed order. C/D require a connected load before sustained boost operation; E may regulate with its switched test load open.

5. For normal shutdown, reduce current demand, inhibit PWM and open B_2. PWM inhibition does not stop passive diode charging, and opening or tripping B_{bp} leaves R_{pc} in circuit without isolating the converter. Measure below 5 V on the DC link and external capacitors before handling; open B_{bp} before restarting.

Faults and return to service

On a fault or unexpected behaviour, inhibit PWM and isolate the source (ITECH output off; B_2 open on C–E). Investigate with the instructor/TA before reset; do not repeatedly reset a trip. Recheck changed wiring or protection settings before energizing. Leave the station discharged and record its condition.

3 What to submit and present

Proposal

Two pages plus figures. The proposal fixes your project and sets the operating points and targets your later work is assessed against. No controller design is required; dynamic targets, control structure and compensator design are presented at the design review. Simulation is encouraged but not required; if you include simulation results, state what the model includes and assumes. Include:

1. Operating envelope and points. The envelope is the range of voltage, current and power (E: reactive power) your station can deliver without exceeding any hardware limit on the station sheet or in the component datasheets. The *binding constraint* is the limit you reach first as you raise power, for example a switch or inductor current rating, the DC voltage limit, the source rating (ITECH 800 W for A/B; transformer 400 VA per phase for C–E) or the load bank. Choose at least two operating points inside the envelope (E: one inductive and one capacitive command) and state each point's *margin* to the relevant limits; for example, 15 A peak against a 20 A rating is a 25% margin. Margin covers ripple, component tolerance, measurement error and overshoot during load changes.

2. Steady-state targets. For each operating point, give measurable values you will check on the bench. Calculate those set by the circuit and operating point, such as voltage, current and power. Targets that depend on how well your controller performs, such as input-current THD and power factor for C and D, are taken from a cited standard, application note or comparable published design, with reasons they suit your station. Targets may be revised later with justification and without penalty.

3. Preliminary switching frequency. Choose a frequency at or below 10 kHz and give your reasons, for example the trade-off between ripple and switching losses, or the harmonic spacing you observed in Lab 2. An estimate of ripple or peak current is welcome but optional: over one switching period the line voltage is nearly constant, so volt-second balance can be applied at the worst-case instant. The frequency is revisited at the design review.

4. Sensing and protection. List each signal you will measure, with its probe, probe setting and ADC scaling (the voltage at the ADC pin at the largest expected value), keeping your trip thresholds inside the ADC range. State your protection thresholds, the overcurrent and overvoltage levels set above your highest operating point and below the hardware limit, and what your firmware does when they trip.

5. Connection diagram. Extend your station-sheet figure, which shows the power circuit only, into the wiring you will build: add each probe and its connection points, the ADC input it feeds, the PWM output driving each switch, the load connections and the instruments.

6. Safety plan. Using §2, write your station's procedures as numbered steps:

- **Startup:** the order of supplies, breakers, load and PWM enable.
- **Normal shutdown:** the reverse sequence, ending with the DC link confirmed below 5 V.
- **Fault response:** what you do if a protection threshold trips or the converter behaves unexpectedly: the control that removes power (ITECH output for A/B, B2 for C–E) and how you confirm the DC link is below 5 V before handling.

7. Work plan. A table of tasks, each with a lead, a supporting member and a target date, through Bench 3. Half a page is enough. Include at least:

- **Before Bench 1:** your switched PLECS model of the station; checks of the provided HIL model's parameters and I/O mapping, and first controller communication (§2); connection diagram finalized.
- **Design review:** small-signal plant model, compensator and dynamic targets (A: control design at the start of Bench 2).
- **Before Bench 2:** firmware and the HIL checklist.
- **Bench 3:** the measurements and who takes them.
- **Report:** who drafts each section.

Plan around the lecture schedule: controller design depends on the dynamics and control lectures, and three-phase work (A) on the three-phase lectures.

8. AI disclosure. Name any AI tools you used and what you used them for, or state that none were used (§4).

Design review

A ten-minute oral review at bench session 1, before powered switching under your firmware. Bring your calculations and models on paper or screen; slides are unnecessary. All members attend, and any member may be asked about any part of the design. Answers must be unaided by AI.

At Bench 1, every group defends operating limits and points, switching frequency, control objectives and structure, dynamic targets for the station-sheet transients, plant model, compensator, predicted stability margins, sensing, protection and bring-up. Any part of the control design not ready at Bench 1 is marked as presented and is reviewed again, unmarked, before that group's closed-loop operation at Bench 2. Project A, in both sections, presents its plant model, compensator and predicted stability margins at the start of Bench 2, where that part is marked, because three-phase control is taught just before Bench 1; the rest of A's review stays at Bench 1.

An unjustified switching frequency or protection setting must be corrected before sign-off. Readiness to energize and the academic mark are separate decisions.

Demonstration and individual questions

Present your design and the evidence you collected at the bench sessions during the registrar-scheduled examination slot; no hardware is operated in the slot. Show your reviewed targets, predicted and measured performance, relevant HIL results, and the conclusions supported by the comparison. All group members contribute. The format, room, group schedule and time allocation will be posted on D2L at least two weeks beforehand.

Each student answers unaided questions about a design decision and a measured result, with follow-ups as needed. Questions may draw on any part of the project. Students receive comparable opportunities to explain their understanding; exhaustive coverage is not expected. The individual mark remains 8% of the course and is excluded from peer adjustment.

Final report

Aim for about 10 pages, excluding appendices. Write for a reader who knows power electronics but has not seen your station. Organize the report around:

- Station, operating points and targets, and system diagram.
- Hardware analysis, switching-frequency choice and control design.
- Simulation and remote HIL validation, including important model limitations.
- Experimental results and a compact table comparing required performance quantities with predictions and targets.
- Interpretation, main loss estimates, efficiency comparison if measured, unresolved questions and proposed improvements.

Label plots with axes, units, operating conditions and probe scaling. Identify the model and firmware versions used for the reported tests. Include an AI disclosure statement. Submit the PLECS models, controller export/settings, relevant HIL model changes, raw data and bench log in a supporting ZIP alongside the PDF; use report appendices for calculations and supporting figures.

4 Assessment and course policies

How marks are assigned

The rubrics on the following pages give the full-marks standard for each criterion. Weights are course percentage points. Apply the scale below separately to each criterion; for example, 80% on a two-point criterion earns 1.6 course points.

Level	Band	Description
Excellent	90–100%	Meets the standard with clear, supported reasoning.
Good	75–89%	Meets most of the standard; minor gaps do not change the conclusions.
Adequate	60–74%	Essential content is present, but reasoning or evidence is limited.
Marginal	50–59%	Significant omissions or errors; the criterion is only partly addressed.
Inadequate	Below 50%	Little usable evidence, or errors that undermine the work.

The marker selects a level and a mark within its band, with a reason, and deducts a weakness under its own criterion; missing data does not exempt dependent analysis. Formatting matters only when it affects interpretation. Supported reasoning can earn full marks even if available evidence cannot resolve a discrepancy.

Group and individual contribution

The proposal, review, group demonstration and report receive group marks. Individual questions receive a separate mark for each student. Every student must understand the complete design, even where implementation work is divided.

Submit one proposal/report PDF per group to the D2L group dropbox by 23:59 on the due date, naming all members and the station. Each student submits confidential peer feedback separately through an individual D2L dropbox. Its form, deadline and instructions will be issued later, sufficiently before submission.

The instructor may adjust an individual student's group-marked subtotal by up to ±4 course points (±10 percentage points of the project), based on justified peer feedback, bench observations and that student's individual answers. The default is zero adjustment. The adjusted subtotal is bounded to 0–32, then the unchanged individual-question mark is added. Example: 24/32 group + 6/8 individual = 30/40; a +4 adjustment for that student gives 34/40.

Use of AI

AI tools are permitted for written work, models, code and study, with disclosure. Name the tools and describe their use in each submission. You are responsible for the correctness of all submitted work. The design review and individual questions assess your own understanding and must be answered without AI assistance. Undisclosed AI use is addressed under the course outline.

Absences and late work

Bench sessions cannot be replaced by another assessment. Late proposals and reports lose 20% per day and are not accepted after three days. The approved course outline governs assessment and absence policies.

5 Proposal and review rubrics

Proposal 4 course points

Criterion	Weight	Full-marks standard
Hardware analysis and operating points	1.5	Derives the operating envelope from the station sheet's hardware limits, identifies the binding constraint, and chooses operating points with stated margins.
Steady-state targets and preliminary switching frequency	1	States measurable steady-state targets for each operating point; calculates those set by the circuit and operating point, and justifies controller-dependent targets from a cited source adapted to the station (the justification is assessed, not the value). Gives a reasoned preliminary frequency at or below 10 kHz; a ripple or peak-current estimate is optional.
Sensing and safety plan	1	Specifies the measured signals, probe settings, ADC scaling and protection thresholds; gives correct station-specific startup, shutdown and fault-response steps, including precharge for C–E.
Work plan	0.5	Gives a task table with leads, support and dates through Bench 3, covering HIL checks, the design review, firmware, measurements and the report, planned around the lecture schedule.

Design review 8 course points

Criterion	Weight	Full-marks standard
Switching frequency and operating limits	2	Defends the operating points and frequency at or below 10 kHz using relevant ratings, losses, delay, bandwidth and resonance; uses ripple calculations only where needed.
Control design	3	States dynamic targets for the station-sheet transients; derives the plant and designs the compensator systematically; predicts crossover and phase margin with implementation delay included; explains limiters and current limiting. A: start of Bench 2.
Sensing and protection	1.5	Explains and demonstrates physical-to-ADC scaling, PWM mapping and programmed protection settings consistent with the design.
Bring-up readiness	1.5	Explains the group's connection diagram, verifies actual wiring, and explains startup, precharge and gates-off conduction before energization. Equivalent understanding is expected of both groups sharing a station.

All members must be able to explain the work. The review is group-marked; the demonstration question period provides the separately assessed individual evidence.

6 Demonstration and report rubrics

Group demonstration 10 course points

Criterion	Weight	Full-marks standard
Experimental evidence	4	Presents evidence of safe, stable closed-loop operation at each operating point and every station-sheet test, captured at the bench sessions. States conditions and signals; compares with targets and required HIL evidence, and quantifies shortfalls. Additional bench transient tests are not required for full marks.
Comparison and explanation	4	Connects predictions to measurements; supports explanations with evidence and identifies unresolved causes and useful further tests.
Communication	2	Presents clear, scaled figures and a coherent technical account within the allotted time; every member contributes.

Individual questions 8 course points

Understanding of the design — 8 points. Explains a design decision and a measured result with correct reasoning, including follow-up questions. Assessed individually using the common quality scale, with comparable opportunities to demonstrate understanding.

Final report 10 course points

Criterion	Weight	Full-marks standard
Hardware analysis	2	Establishes operating limits, current margins and sensing requirements and uses them in design decisions. Explains AC-filter role and relevant resonance; derives ripple only where needed for the topology.
Control and model validation	3	Presents models, compensators and predicted responses; C–E show synchronization lock and explain how the provided PLL model is used; compares linearised, switched and HIL results and identifies model limitations.
Experimental evidence and comparison	3	Provides the station-sheet waveforms and metrics at each operating point, with conditions and scaling, compared against predictions and targets. Includes every station-sheet transient. Additional bench operating points or transients are not required for full marks.
Interpretation and losses	2	Uses evidence to discuss discrepancies and uncertainty; estimates main losses, explains operating/thermal implications and proposes justified improvements. Compares efficiency if measured.

Teamwork is evidenced by the work plan, peer assessment and bench observations. The station sheet defines the converter function, hardware and bench demonstration. These rubrics apply to all groups; additional tests are not required for full marks.